

Ridge, Lasso and OLS comparison with simulated data in Python

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Introduction

Compare the performance of regularized and standard linear regression models in a high-dimensional environment

- Simulate data from an high-dimensional DGP
- Use K-Fold Cross-Validation in order to train and test different linear regression machine learning models
 - OLS
 - Ridge
 - Lasso
- Check and compare the different results in terms of MSE

Why Regularization

- **Bias-Variance tradeoff**: adding too many regressors could trade off in fitting the noise but not the DGP, leading to poor out of sample performance because of the high variance of the model
- **($n < p$)**: too many predictors with respect to observations bringing to unstable estimates and high variance
- **Multicollinearity**: two or more regressors are correlated among them, so the estimates are very unstable

The models

- OLS:

$$\hat{\beta}^{\text{OLS}} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \sum_{j=1}^p x_{ij} \beta_j \right)^2 \right\}$$

- Ridge:

$$\hat{\beta}^{\text{ridge}} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

- Lasso:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Data simulation

Each dataset of 2400 observations and 600 regressors is simulated from the following DGP:

$$y_i = 5 + X_i\beta + \varepsilon_i$$

- The beta coefficients are simulated from a standard normal distribution:
 $\beta \sim \mathcal{N}(0, 1)$
- The value of the regressors is simulated for each observation i from

$$X_{ij} \sim \mathcal{N}(0, sd_j^2)$$

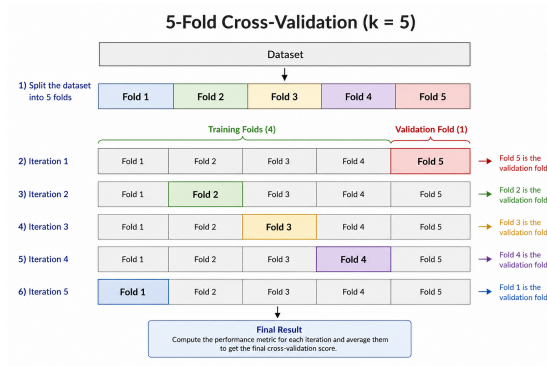
where $sd_j \sim \chi^2(3)$

- The heteroskedastic errors ε_i are drawn from: $\varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$
- 600 other variables that are not relevant for the DGP are stacked to the dataset in order to test the ability of the regularized models to shrink their coefficients

K-Fold Cross Validation

In order to choose the optimal hyperparameter α , the model is estimated using a K-Fold Cross Validation approach with $k = 5$, then the optimal α is chosen as the one that minimizes the MSE on average among the different validation sets:

$$\alpha^* = \arg \min_{\alpha} \left(\frac{1}{k} \sum_{j=1}^k MSE_j(\alpha) \right)$$



Optimal α

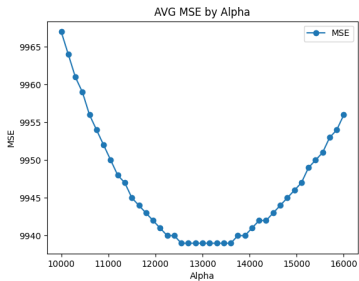


Figure: Ridge loss function

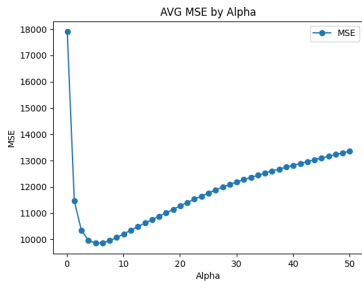


Figure: Lasso loss function

Comparative results

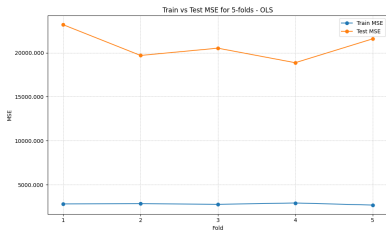


Figure: OLS

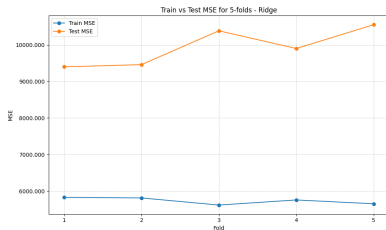


Figure: Ridge

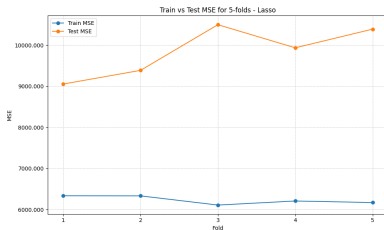


Figure: Lasso

Python libraries

- NumPy in order to draw from the standard normal and chi-square distributions
- Pandas in order to manage and manipulate data
- Scikit-learn in order to estimate the OLS, Ridge and Lasso models
- Matplotlib for graphs

Final considerations

- Ridge and Lasso outperform the OLS as expected
- A possible extension could introduce some multicollinearity of the predictors in order to test how regularization helps